

## **Practice Question Set :**

- 1. What are the four basic operations performed by a computer?**
  - 2. Which language does a computer understand directly?**
  - 3. What are mnemonics in Assembly Language?**
  - 4. What is a compiler?**
  - 5. Who developed C language and in which year?**
  - 6. Why was C language developed?**
  - 7. Why is C called a middle-level language?**
  - 8. Write any four features of C language.**
  - 9. What is portability?**
  - 10. What does `stdio.h` stand for?**
  - 11. Why is machine language difficult to use?**
  - 12. Why is Assembly Language machine dependent?**
  - 13. Why were high-level languages introduced?**
  - 14. Explain the compilation process.**
  - 15. Explain the history of C language.**
  - 16. Explain the importance of rewriting Unix in C.**
  - 17. Why was `gets()` removed from C11?**
  - 18. What is the purpose of the `main()` function?**
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# Important Interview/Viva Questions

**19. Why is C still popular today?**

**20. Difference between Compiler and Interpreter.**

**21. Difference between Machine Language and Assembly Language.**

**22. Difference between Assembly Language and High-Level Language.**

**23. Why is C considered fast?**

**24. What happens if main() is not present in a C program?**

**25. Why is fgets() preferred over gets()?**

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## Answers :

**1. What are the four basic operations performed by a computer?**

**Answer:**

- Input
- Processing
- Storage
- Output

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**2. Which language does a computer understand directly?**

**Answer:**

Computers understand only **Machine Language (Binary Language)** consisting of 0s and 1s.

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**3. What are mnemonics in Assembly Language?**

**Answer:**

Mnemonics are symbolic instructions used instead of binary codes.

Examples:

- MOV
  - ADD
  - SUB
- 

## 4. What is a compiler?

**Answer:**

A compiler is a software that translates a high-level language program into machine language.

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## 5. Who developed C language and in which year?

**Answer:**

C language was developed by **Dennis Ritchie** in **1972** at **Bell Labs**.

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## 6. Why was C language developed?

**Answer:**

C was developed to overcome the limitations of Assembly Language and B Language and to rewrite the Unix operating system.

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## 7. Why is C called a middle-level language?

**Answer:**

C combines features of both high-level and low-level languages. It provides easy syntax as well as direct memory access through pointers.

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## 8. Write any four features of C language.

**Answer:**

- Simple
- Fast
- Portable

- Structured
  - Rich library functions
  - Efficient memory usage
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## 9. What is portability?

**Answer:**

Portability means a program written on one computer can run on another computer with little or no modification.

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## 10. What does stdio.h stand for?

**Answer:**

stdio.h stands for **Standard Input Output Header File**.

It provides functions like:

- printf()
  - scanf()
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## 11. Why is machine language difficult to use?

**Answer:**

Machine language consists of binary numbers (0 and 1), making it difficult to write, understand, debug, and maintain.

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## 12. Why is Assembly Language machine dependent?

**Answer:**

Assembly Language instructions vary from processor to processor, so programs written for one processor may not work on another.

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## 13. Why were high-level languages introduced?

**Answer:**

High-level languages were introduced to make programming easier, more readable, and less error-prone.

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## 14. Explain the compilation process.

**Answer:**

Source Code  
↓  
Compiler  
↓  
Machine Code  
↓  
Execution  
↓  
Output

The compiler converts the source code into machine code, which is then executed by the computer.

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## 15. Explain the history of C language.

**Answer:**

Machine Language  
↓  
Assembly Language  
↓  
BCPL (1966) - Martin Richards  
↓  
B Language (1969) - Ken Thompson  
↓  
C Language (1972) - Dennis Ritchie

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## 16. Explain the importance of rewriting Unix in C.

**Answer:**

Rewriting Unix in C made it portable, efficient, and easier to maintain. It allowed Unix to run on different hardware platforms.

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## 17. Why was gets() removed from C11?

**Answer:**

gets() was removed because it does not perform bounds checking and may cause buffer overflow. fgets() is safer because it limits the number of characters read.

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## 18. What is the purpose of the main() function?

**Answer:**

main() is the starting point of every C program. Program execution begins from this function.

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## 19. Why is C still popular today?

**Answer:**

Because C is:

- Fast
  - Efficient
  - Portable
  - Structured
  - Used in operating systems, embedded systems, and compilers
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## 20. Difference between Compiler and Interpreter

| Compiler                         | Interpreter              |
|----------------------------------|--------------------------|
| Translates whole program at once | Translates line by line  |
| Faster execution                 | Slower execution         |
| Generates object code            | No object code generated |
| Example: C                       | Example: Python          |

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## 21. Difference between Machine Language and Assembly Language

| Machine Language           | Assembly Language    |
|----------------------------|----------------------|
| Uses 0s and 1s             | Uses mnemonics       |
| Difficult to understand    | Easier to understand |
| Directly understood by CPU | Requires assembler   |

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## 22. Difference between Assembly Language and High-Level Language

**Assembly Language      High-Level Language**

|                    |                     |
|--------------------|---------------------|
| Machine dependent  | Machine independent |
| Hard to write      | Easy to write       |
| Less readable      | More readable       |
| Requires assembler | Requires compiler   |

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## 23. Why is C considered fast?

**Answer:**

C is close to hardware and produces efficient machine code with minimal overhead, resulting in faster execution.

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## 24. What happens if `main()` is not present in a C program?

**Answer:**

Without the `main()` function, the program cannot start execution and will produce an error.

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## 25. Why is `fgets()` preferred over `gets()`?

**Answer:**

`fgets()` allows specifying the maximum number of characters to read, preventing buffer overflow and making it safer than `gets()`.